

LangChain Overview

LangChain is an open-source framework for building applications powered by Large Language Models (LLMs). It helps developers connect LLMs with external data sources, APIs, tools, and databases. Core Components: Document Loaders Text Splitters Embedding Models Vector Stores Retrievers Prompt Templates Chains Agents Memory Output Parsers A typical RAG pipeline is: Documents → Loader → Splitter → Embeddings → Vector Store → Retriever → Prompt → LLM → Final Answer.

Document Loaders in LangChain

Document Loaders are responsible for reading data from different file formats and converting them into Document objects. Common loaders: TextLoader (.txt) PyPDFLoader (.pdf) CSVLoader (.csv) Docx2txtLoader (.docx) UnstructuredHTMLLoader (.html) Each Document contains: **page_content**: extracted text **metadata**: source, page number, author, title, etc. PyPDFLoader returns one Document object per PDF page, making it ideal for chunking and Retrieval-Augmented Generation workflows.